

Report: Group Policy Management

1. Introduction

Three separate group policies were created and applied using the Group Policy Management Console on the domain controller. These policies were configured to enforce a 15-minute screen lock timeout, disable SMBv1 protocol across all domain machines, and apply a logon banner before login. Each GPO was created under Group Policy Objects, configured with the required settings, and linked to the Finance OU, which contains the PC1 machine. The policies were updated on the client using `gpupdate /force`, and the results were verified directly through system response and command-line output on PC1.

2. Objective

The objective of these tasks was to apply specific security and usability settings to domain-joined machines using Group Policy. The first task aimed to configure automatic screen lock after 15 minutes of inactivity. The second task involved disabling the SMBv1 protocol on all client machines in the domain. The third task involved applying a logon banner message that would display a legal notice before the user could log in. Each policy was applied to the Finance OU, and tested using PC1 to confirm that the intended setting was working.

3. Tools used

- Windows Server (AD DS)
- Group Policy Management Console (GPMC)
- Windows PowerShell
- Windows 10 Enterprise (PC1)

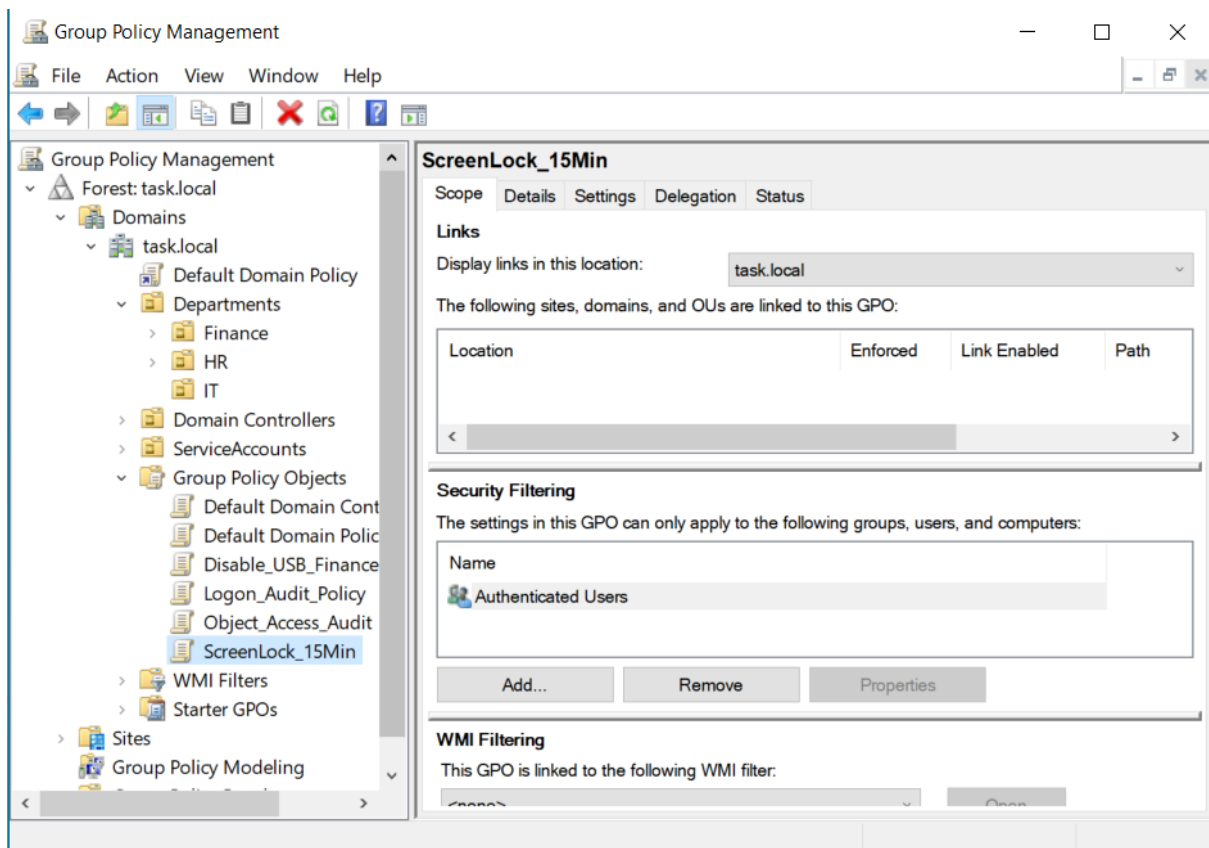
4. Methodology

4.1 Enforce 15-Minute Screen Lock via GPO

A new GPO named ScreenLock_15Min was created. In the GPO settings, the following were configured under Computer Configuration:

- “Turn off the display (plugged in)” was set to 15 minutes
- “Interactive logon: Machine inactivity limit” was set to 900 seconds

The GPO was linked to the Finance OU where PC1 is located. After running gpupdate /force on PC1, the screen lock was verified by waiting 15 minutes. The system locked automatically, requiring the user to re-enter their password.



(Create a new GPO ScreenLock_15min)

Screen saver timeout

Screen saver timeout

Previous Setting Next Setting

☐ Not Configured Comment:

☒ Enabled

☐ Disabled

Supported on: At least Windows 2000 Service Pack 1

Options:

Number of seconds to wait to enable the screen saver

Seconds: 900

Help:

Specifies how much user idle time must elapse before the screen saver is launched.

When configured, this idle time can be set from a minimum of 1 second to a maximum of 86,400 seconds, or 24 hours. If set to zero, the screen saver will not be started.

This setting has no effect under any of the following circumstances:

- The setting is disabled or not configured.
- The wait time is set to zero.
- The "Enable Screen Saver" setting is disabled.
- Neither the "Screen saver executable name" setting nor the Screen Saver dialog of the client computer's Personalization or Display Control Panel specifies a valid existing screen saver program on the client.

OK Cancel Apply

(Edited the policy and now timeout is enabled)

Windows Firewall Control Panel - Password protect the screen saver

Previous Setting Next Setting

☐ Not Configured Comment:

☒ Enabled

☐ Disabled

Supported on:

Options:

Help:

Determines whether screen savers used on the computer are password protected.

If you enable this setting, all screen savers are password protected. If you disable this setting, password protection cannot be set on any screen saver.

This setting also disables the "Password protected" checkbox on the Screen Saver dialog in the Personalization or Display Control Panel, preventing users from changing the password protection setting.

If you do not configure this setting, users can choose whether or not to set password protection on each screen saver.

To ensure that a computer will be password protected, enable the "Enable Screen Saver" setting and specify a timeout via the "Screen Saver timeout" setting.

Note: To remove the Screen Saver dialog, use the "Prevent

OK Cancel Apply

(screen savers used on the computer are password protected)

Enable screen saver

Enable screen saver

Previous Setting Next Setting

☐ Not Configured Comment:

☒ Enabled

☐ Disabled

Supported on: At least Windows 2000 Service Pack 1

Options:

Help:

Enables desktop screen savers.

If you disable this setting, screen savers do not run. Also, this setting disables the Screen Saver section of the Screen Saver dialog in the Personalization or Display Control Panel. As a result, users cannot change the screen saver options.

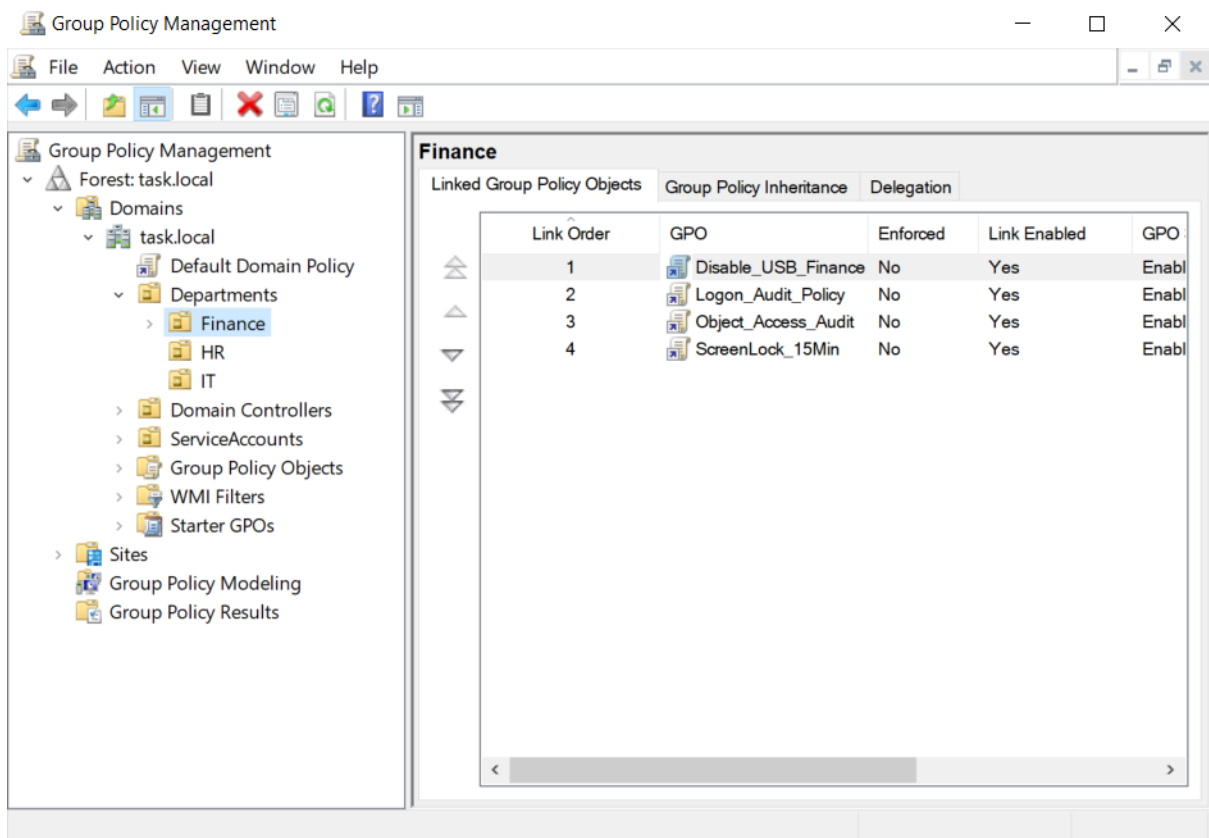
If you do not configure it, this setting has no effect on the system.

If you enable it, a screen saver runs, provided the following two conditions hold: First, a valid screen saver on the client is specified through the "Screen Saver executable name" setting or through Control Panel on the client computer. Second, the screen saver timeout is set to a nonzero value through the setting or Control Panel.

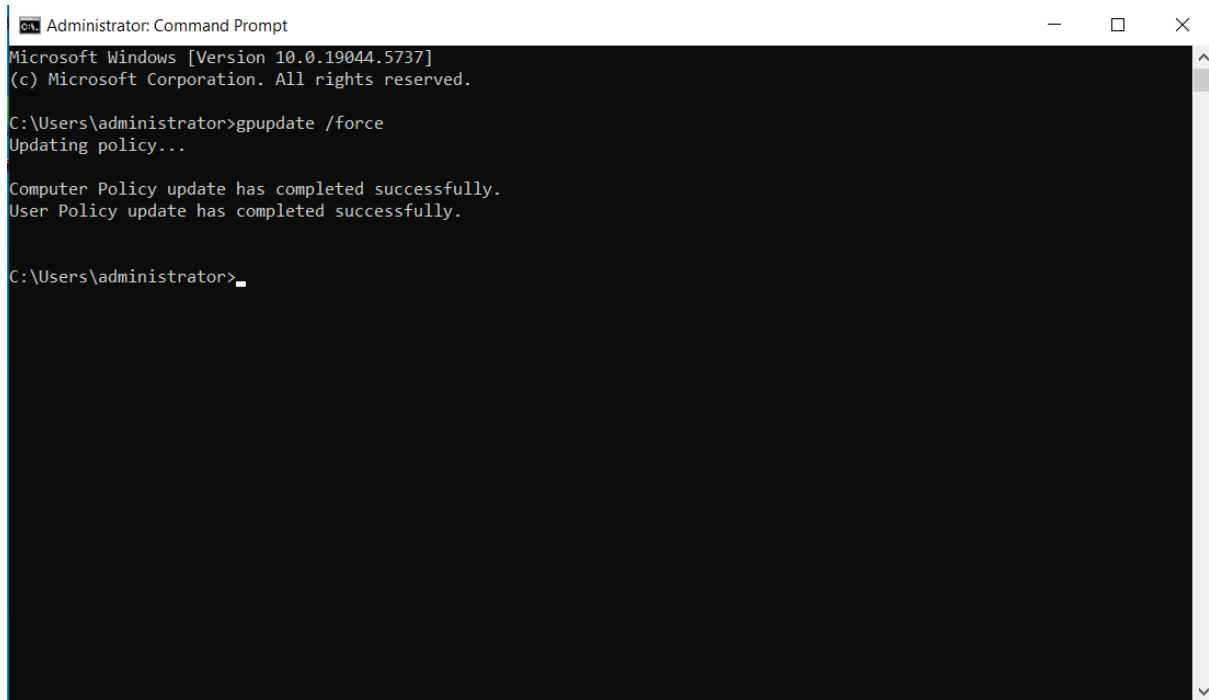
Also, see the "Prevent changing Screen Saver" setting.

OK Cancel Apply

(Enables desktop screen savers)



(Link new GPO with Finance because our PC1 is present here)



(Policy was updated)

```
C:\> Administrator: Command Prompt
Microsoft Windows [Version 10.0.19044.5737]
(c) Microsoft Corporation. All rights reserved.

C:\Users\administrator>gpresult /r

Microsoft (R) Windows (R) Operating System Group Policy Result tool v2.0
© Microsoft Corporation. All rights reserved.

Created on 5/12/2025 at 11:06:44 AM

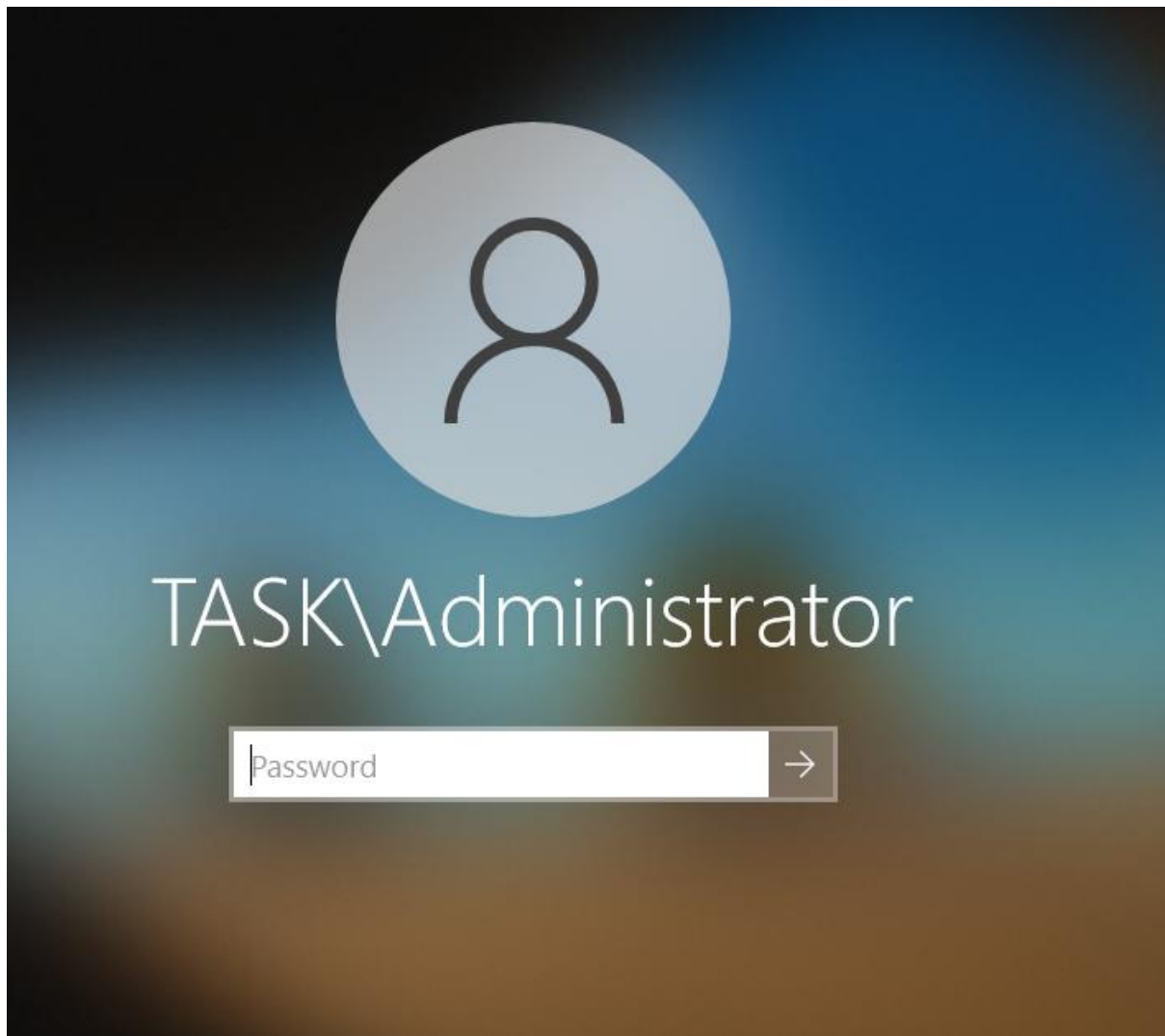
RSOP data for TASK\Administrator on PC1 : Logging Mode
-----

OS Configuration:      Member Workstation
OS Version:            10.0.19044
Site Name:             Default-First-Site-Name
Roaming Profile:       N/A
Local Profile:         C:\Users\administrator
Connected over a slow link?: No

COMPUTER SETTINGS
-----
CN=PC1,OU=Finance,OU=Departments,DC=task,DC=local
Last time Group Policy was applied: 5/12/2025 at 11:05:56 AM
Group Policy was applied from:      server1.task.local
Group Policy slow link threshold:   500 kbps
Domain Name:                       TASK
Domain Type:                       Windows 2008 or later

Applied Group Policy Objects
-----
  Disable_USB_Finance
  Logon_Audit_Policy
  Object_Access_Audit
  ScreenLock_15Min
  Default Domain Policy
```

(Applied successfully to PC1)



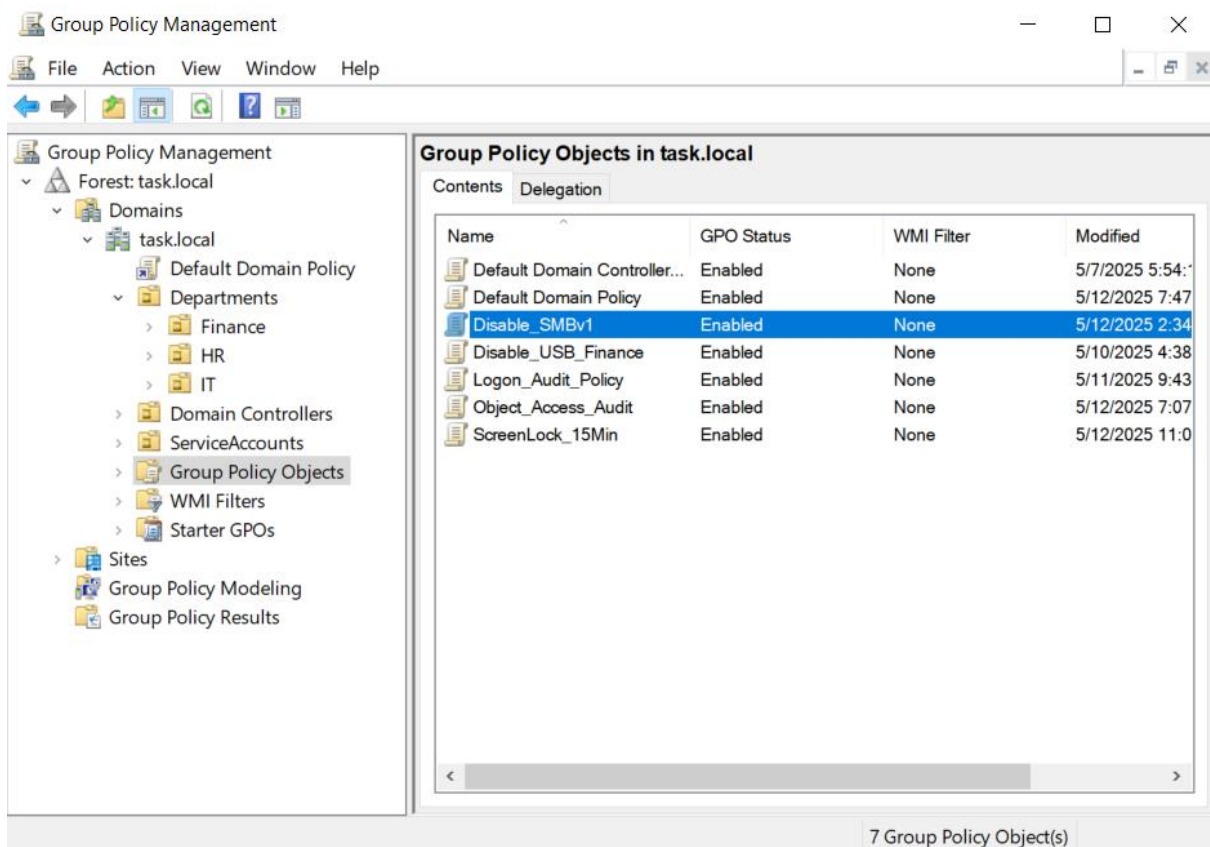
(Screen saver with password)

4.2 Disable SMBv1 Across the Domain

A GPO named Disable_SMBv1 was created. Under Computer Configuration → Preferences → Windows Settings → Registry, a new registry item was added:

- Hive: HKEY_LOCAL_MACHINE
- Key: SYSTEM\CurrentControlSet\Services\LanmanServer\Parameters
- Name: SMB1, Type: REG_DWORD, Data: 0

The GPO was linked to the Finance OU. On PC1, gpupdate /force was executed, and PowerShell command Get-WindowsOptionalFeature -Online -FeatureName SMB1Protocol confirmed the feature was in State : Disabled.



(create a new GPO Disable_SMBv1)

SMB1 Properties

General Common

 Action: Update

Hive: HKEY_LOCAL_MACHINE

Key Path: SYSTEM\CurrentControlSet\Services\LanmanSe ...

Value name

☐ Default SMB1

Value type: REG_DWORD

Value data: 00000000

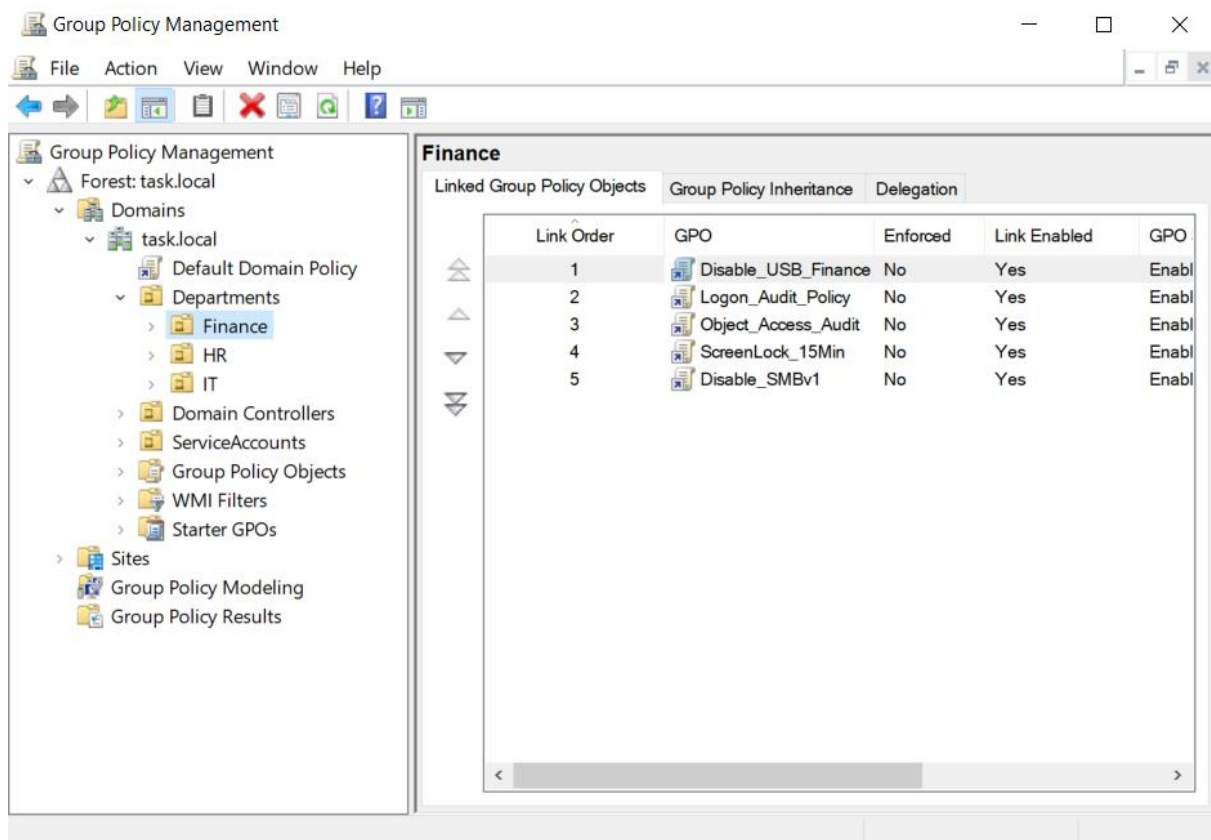
Base

☒ Hexadecimal

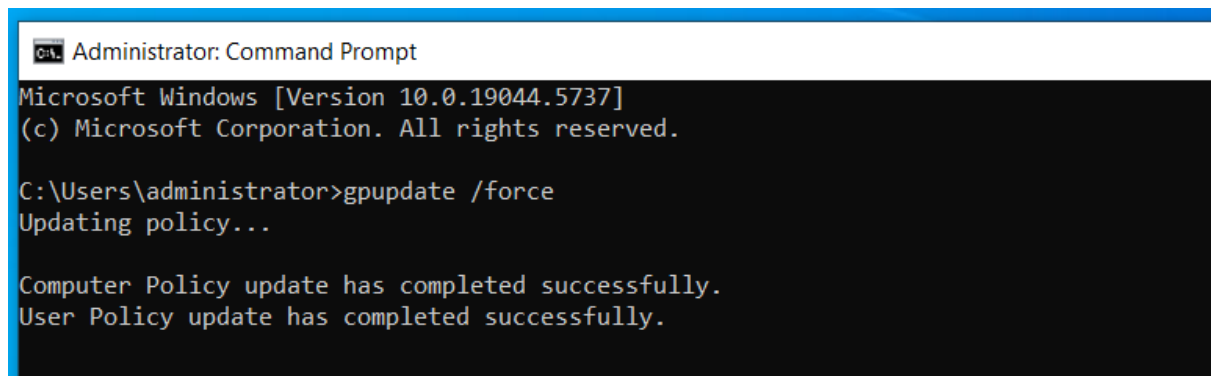
☐ Decimal

OK Cancel Apply Help

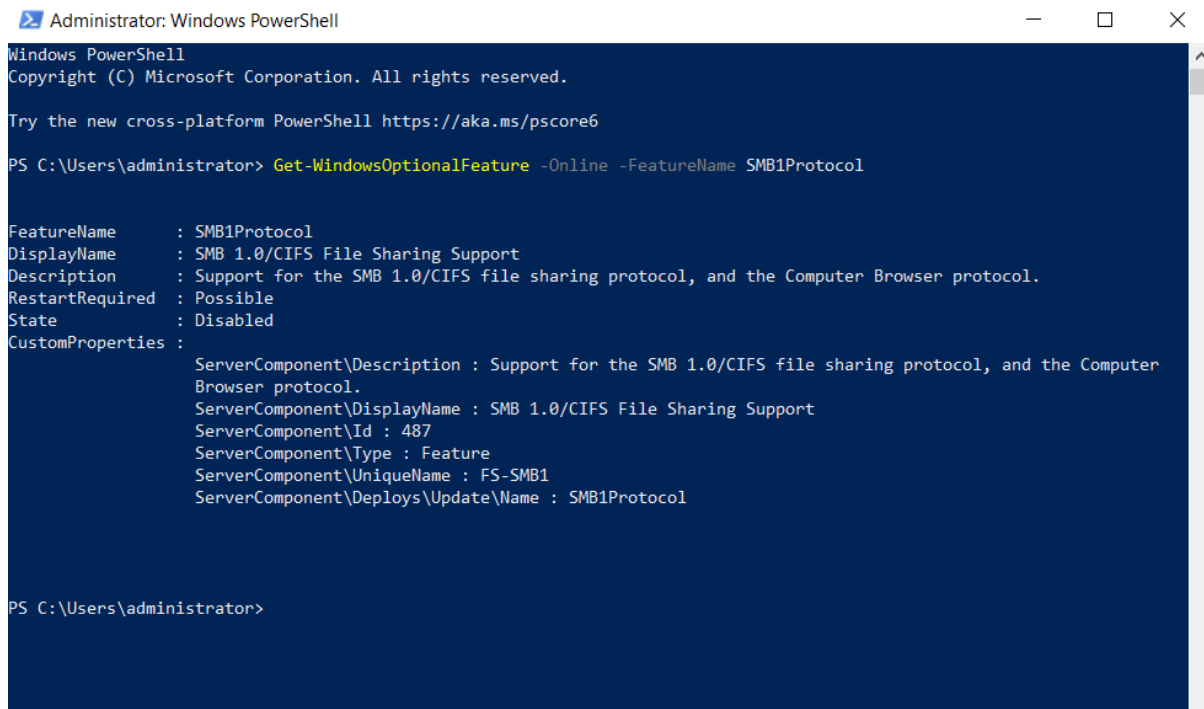
(edit policy)



(Link GPO with Finance)



(update policy)



```
Administrator: Windows PowerShell
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Try the new cross-platform PowerShell https://aka.ms/pscore6

PS C:\Users\administrator> Get-WindowsOptionalFeature -Online -FeatureName SMB1Protocol

FeatureName      : SMB1Protocol
DisplayName       : SMB 1.0/CIFS File Sharing Support
Description      : Support for the SMB 1.0/CIFS file sharing protocol, and the Computer Browser protocol.
RestartRequired  : Possible
State            : Disabled
CustomProperties :
                  ServerComponent\Description : Support for the SMB 1.0/CIFS file sharing protocol, and the Computer
                  Browser protocol.
                  ServerComponent\DisplayName : SMB 1.0/CIFS File Sharing Support
                  ServerComponent\Id         : 487
                  ServerComponent\Type       : Feature
                  ServerComponent\UniqueName : FS-SMB1
                  ServerComponent\Deploys\Update\Name : SMB1Protocol

PS C:\Users\administrator>
```

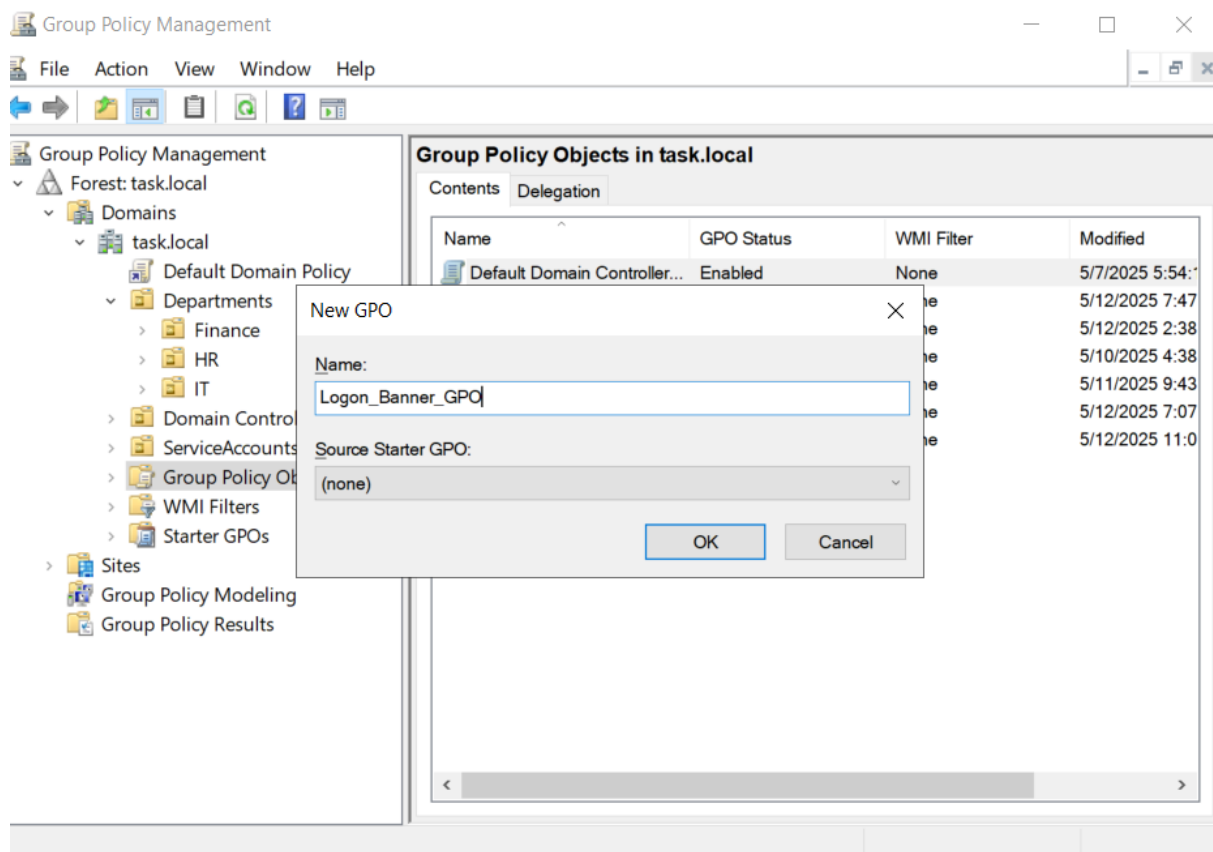
(SMBv1 is disabled)

4.3 Deploy Logon Banner via GPO

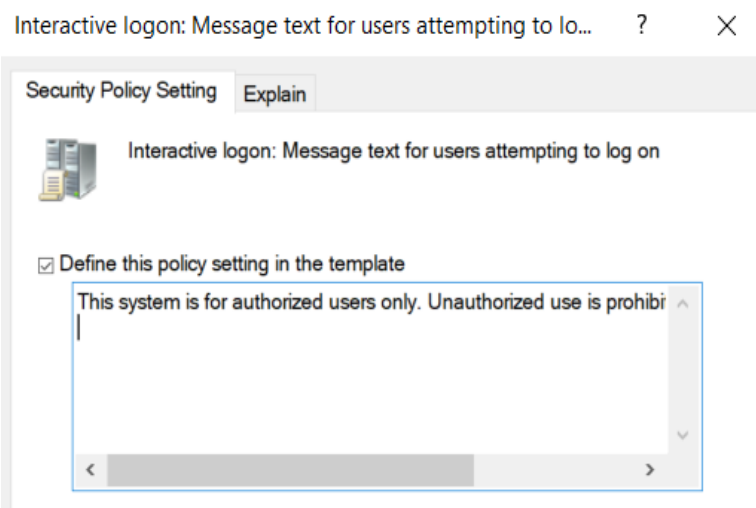
A GPO named Logon_Banner_GPO was created. The following two settings were configured:

- “Interactive logon: Message title for users attempting to log on” was set to: Authorized Access Only
- “Interactive logon: Message text for users attempting to log on” was set to: This system is for authorized users only. Unauthorized use is prohibited and subject to monitoring.

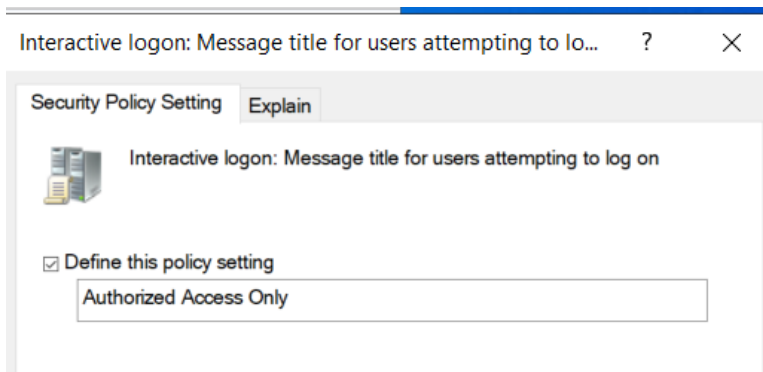
The GPO was linked to the Finance OU. On PC1, after a policy update and reboot, the login banner appeared before the username and password prompt on the login screen.



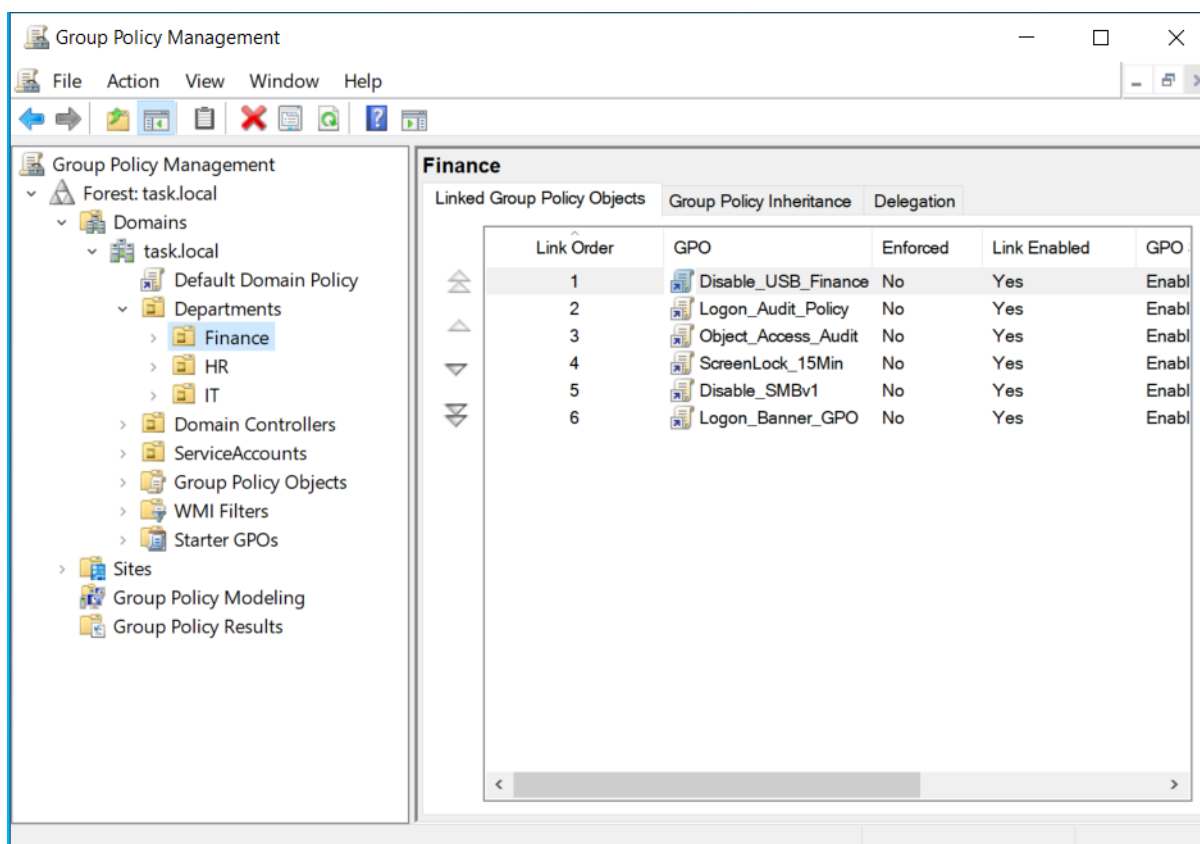
(create a new GPO – Logon_Banner_GPO)



(edit policy)



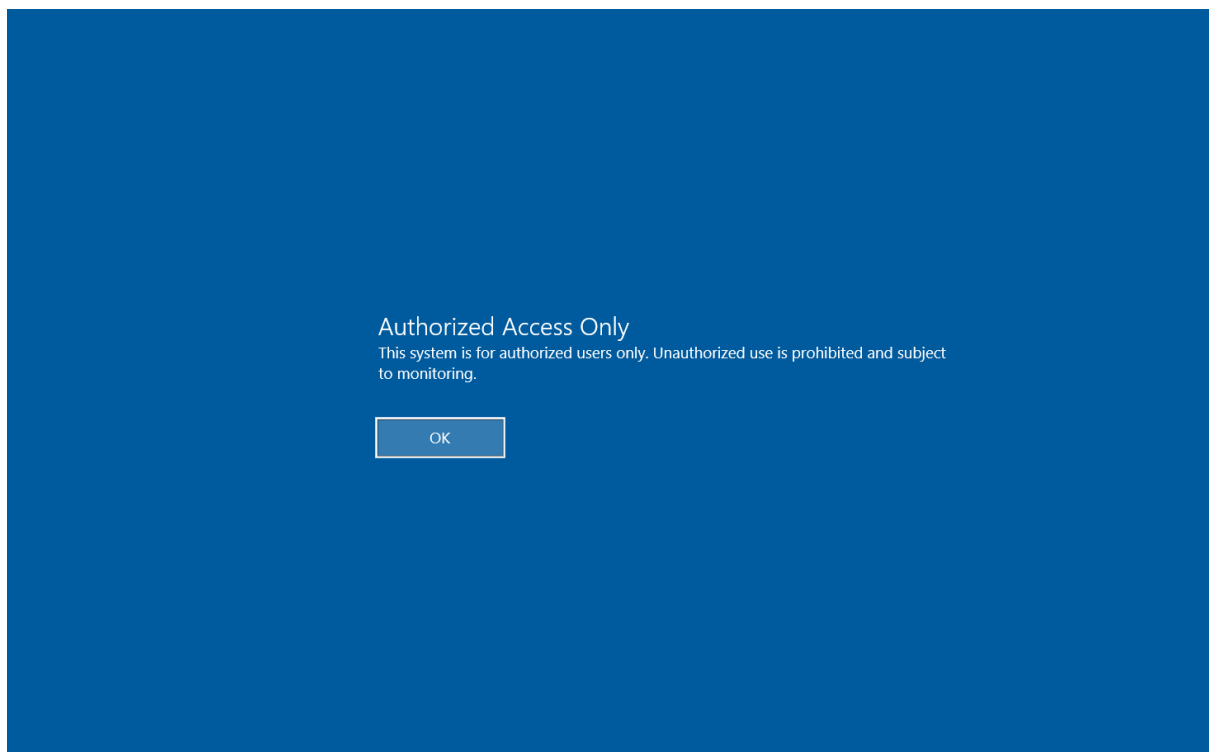
(edit policy)



(link GPO with finance)

```
(c) Microsoft Corporation. All rights reserved.  
  
C:\Users\administrator>gpupdate /force  
Updating policy...  
  
Computer Policy update has completed successfully.  
User Policy update has completed successfully.
```

(Policy was updated)



(Policy is working perfectly)

5. Recommendations

- Apply screen lock settings under Computer Configuration to ensure consistent enforcement across all machines.
- Use the registry method to disable SMBv1 when managing older operating systems that do not support policy-based feature removal.
- Always confirm that the GPO is linked to the correct OU that contains the computer object, not just the user.

- Reboot machines if a policy does not apply after running gpupdate.
- Regularly verify applied GPOs using gpresult /r to confirm correct linkage and policy application.

6. Conclusion

Three separate GPOs were created and successfully applied. The first enforced a 15-minute screen lock after user inactivity, which was confirmed by idle session lock on PC1. The second GPO disabled SMBv1 via a registry-based policy, and the feature was verified as disabled through PowerShell output. The third GPO introduced a legal warning banner at the login screen, which appeared as expected for users of PC1. All group policy objectives were completed, confirmed, and applied as per requirements.